

| FTM 1 MODULE OBJECTIVES |            |                                                               |                                 |                                                                                                                                                                                                              |
|-------------------------|------------|---------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activity                | Discipline | Title                                                         | OBJECTIVE CODE                  | Lecture Objective                                                                                                                                                                                            |
| DLA 1                   | BCHM; HCB  | Membrane Structure and Transport: Integrated Basic Principles | SOM.MK.1.BPM1.1.FTM.3.HCB.o683  | Differentiate between the two (2) major classes of membrane proteins.                                                                                                                                        |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.o684  | List the two major categories of integral membrane proteins.                                                                                                                                                 |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.o685  | Describe the six major functions of the two (2) major categories of integral membrane proteins.                                                                                                              |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.o686  | Discuss membrane permeability in relation to membrane composition, and hydrophilic and lipophilic compounds                                                                                                  |
|                         |            |                                                               | SOM.MK.I.BPM1.1.FTM.3.BCHM.0010 | Describe the structure, composition and significance of the fluid mosaic model of cell membrane                                                                                                              |
|                         |            |                                                               | SOM.MK.I.BPM1.1.FTM.3.BCHM.0011 | Review the lipid (cholesterol, phospholipids and glycolipids) composition of the plasma membrane. Discuss the relevance of cholesterol and lipid composition in regulating membrane fluidity                 |
|                         |            |                                                               | SOM.MK.I.BPM1.1.FTM.3.BCHM.0014 | Differentiate between active and passive transport systems based on energy requirement and movement along the concentration gradient.                                                                        |
|                         |            |                                                               | SOM.MK.I.BPM1.1.FTM.3.BCHM.0015 | Discuss facilitated diffusion by glucose transporters (GLUT 1-5) highlighting their location and characteristics in relation to substrate affinity and insulin sensitivity.                                  |
|                         |            |                                                               | SOM.MK.I.BPM1.1.FTM.3.BCHM.0017 | Distinguish between primary active transport (Na <sup>+</sup> /K <sup>+</sup> -ATPase) and secondary active transport (SGLT).                                                                                |
|                         |            |                                                               | SOM.MK.I.BPM1.1.FTM.3.BCHM.0018 | Describe the CFTR channel as an example of ABC transporters. Predict effects of CFTR mutations on lung and pancreas.                                                                                         |
| Lecture 1               | HCB        | Histology of the Cytoskeleton of the Cell                     | SOM.MK.1.BPM1.1.FTM.3.HCB.0709  | List the three domains of cellular life.                                                                                                                                                                     |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0710  | List the three major types of protein filaments (cytoskeletal elements) that form the cytoskeleton.                                                                                                          |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0711  | State the function(s) of microtubules as a cytoskeletal element.                                                                                                                                             |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0712  | State the function(s) of intermediate filaments as a cytoskeletal element.                                                                                                                                   |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0713  | State the function(s) of microfilaments as a cytoskeletal element.                                                                                                                                           |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0714  | Describe the protein structure of microtubules as a cytoskeletal element.                                                                                                                                    |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0715  | Describe the protein structure of intermediate filaments as a cytoskeletal element.                                                                                                                          |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0716  | Describe the protein structure of microfilaments as a cytoskeletal element.                                                                                                                                  |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0717  | Describe the assembly of microtubules.                                                                                                                                                                       |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0718  | Describe the assembly of intermediate filaments.                                                                                                                                                             |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0719  | Describe the assembly of microfilaments.                                                                                                                                                                     |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0720  | Discuss three compounds that can affect microtubule function.                                                                                                                                                |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0721  | Discuss two compounds that can affect microfilament function.                                                                                                                                                |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0722  | State the function(s) of the centrosome.                                                                                                                                                                     |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0723  | Describe the structure of the centrosome.                                                                                                                                                                    |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0724  | Give the location of the basal body.                                                                                                                                                                         |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0725  | State the function(s) of the basal body.                                                                                                                                                                     |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0726  | Describe the structure of the basal body.                                                                                                                                                                    |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0727  | Differentiate between the following apical specializations based on function, tissues found and their microanatomical structures: primary cilia, motile cilia and flagella.                                  |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0728  | Differentiate between the following apical specializations based on function and their microanatomical structures: microvilli, stereocilia.                                                                  |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0729  | Describe the terminal web.                                                                                                                                                                                   |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0730  | Describe the motor protein associated with microfilaments.                                                                                                                                                   |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0731  | Describe the two motor proteins associated with microtubules.                                                                                                                                                |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0732  | Describe role of the cytoskeleton in intracellular transport.                                                                                                                                                |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0733  | Define actin polymerization.                                                                                                                                                                                 |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0734  | Describe the three types of membrane protrusion structures.                                                                                                                                                  |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0735  | Describe the role of actin polymerization in membrane protrusion.                                                                                                                                            |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0736  | Describe the role of actin polymerization in cellular motility.                                                                                                                                              |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0737  | Explain how a cell moves across a substrate by utilizing the concept of actin polymerization.                                                                                                                |
| Lecture 2               | HCB        | Histology of Cellular Organelles                              | SOM.MK.1.BPM1.1.FTM.3.HCB.0738  | Discuss the nucleus as a major organelle.                                                                                                                                                                    |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0739  | Give the function(s) of the nucleolus.                                                                                                                                                                       |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0740  | Describe the microscopic structure of the nucleolus.                                                                                                                                                         |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0741  | State the three components of chromatin.                                                                                                                                                                     |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0742  | Describe how these three components come together to form chromatin.                                                                                                                                         |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0743  | Distinguish between euchromatin and heterochromatin in a nucleus.                                                                                                                                            |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0744  | Predict cellular activity based on the nuclear chromatin structure.                                                                                                                                          |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0745  | Give the function of the nuclear envelope.                                                                                                                                                                   |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0746  | Describe the structure of nuclear envelope including the nuclear pore complex.                                                                                                                               |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0747  | Differentiate between the following organelles based on their microanatomical structures and function: ribosomes, endoplasmic reticulum (RER & SER), Golgi apparatus, lysosome, peroxisome and mitochondria. |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0748  | Describe the bi-directional vesicle transport between the endoplasmic reticulum and Golgi apparatus.                                                                                                         |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0749  | Describe protein trafficking for lysosomes, secretion, plasma membrane, nucleus, mitochondria, cytosol and peroxisomes.                                                                                      |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0750  | Describe the three cellular pathways to lysosomal degradation.                                                                                                                                               |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0751  | Identify the consequences of dysfunctional peroxisomes for example Zellweger syndrome.                                                                                                                       |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0752  | Identify the causes and consequences of lysosomal storage diseases (for example Tay-Sachs) based on the defective enzyme, accumulated products and organs most susceptible.                                  |
|                         |            |                                                               | SOM.MK.1.BPM1.1.FTM.3.HCB.0753  | Define autophagy                                                                                                                                                                                             |

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|-----------|------------|-----------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.HCB.0754   | Describe the process of autophagy including the coordinated process from the membrane conformational changes to the fusion with lysosomes (autolysosomes).                                                                                                                                                                                                                                                                              |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.HCB.0755   | Discuss the significance of autophagy                                                                                                                                                                                                                                                                                                                                                                                                   |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.HCB.0756   | Give the structure and function (s) of the proteasome                                                                                                                                                                                                                                                                                                                                                                                   |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.HCB.0757   | Describe the structure of the proteasome.                                                                                                                                                                                                                                                                                                                                                                                               |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.HCB.0758   | Describe the five types of intracellular inclusions.                                                                                                                                                                                                                                                                                                                                                                                    |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.HCB.0759   | Describe the four basic tissue types.                                                                                                                                                                                                                                                                                                                                                                                                   |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.HCB.0760   | Identify the origin of the four basic types of tissue.                                                                                                                                                                                                                                                                                                                                                                                  |
| DLA 2     | HCB        | Cellular Biology of Vesicular Transport | SOM.MK.1.BPM1.1.FTM.3.HCB.0687   | List the two major types of vesicular transport.                                                                                                                                                                                                                                                                                                                                                                                        |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.HCB.0688   | Describe the precise targeting of vesicles within the cell.                                                                                                                                                                                                                                                                                                                                                                             |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.HCB.0689   | Describe the three different mechanisms of endocytosis.                                                                                                                                                                                                                                                                                                                                                                                 |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.HCB.0690   | Describe the role of endosomes in vesicular transport.                                                                                                                                                                                                                                                                                                                                                                                  |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.HCB.0691   | Describe the four pathways for processing internalized ligand-receptor complexes.                                                                                                                                                                                                                                                                                                                                                       |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.HCB.0692   | Describe the two general pathways of exocytosis.                                                                                                                                                                                                                                                                                                                                                                                        |
| DLA 3     | BCHM       | Structure and Function of Lipids        | SOM.MK.1.BPM1.1.FTM.3.BCHM.0001  | Review the grouping of lipids (Fatty acids, triacylglycerols, phospholipids, sphingolipids, steroids)                                                                                                                                                                                                                                                                                                                                   |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.BCHM.0002  | Review the significance and functions of cholesterol in humans.                                                                                                                                                                                                                                                                                                                                                                         |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.BCHM.0003  | Describe fatty acid structure. Discuss the relationship between chain length and desaturation and relate its significance to fluidity of cell membrane                                                                                                                                                                                                                                                                                  |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.BCHM.0004  | Discuss importance of dietary essential fatty acids: linoleic acid (omega-6) and alpha-linolenic acid (omega-3)                                                                                                                                                                                                                                                                                                                         |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.BCHM.0005  | Review the grouping of fatty acids into omega-6 and omega-3 families and describe the synthesis of arachidonic acid and of docosahexaenoic acid (DHA)                                                                                                                                                                                                                                                                                   |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.BCHM.0006  | Describe structure-function relationship of triacylglycerols.                                                                                                                                                                                                                                                                                                                                                                           |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.BCHM.0007  | Distinguish between phospholipids (Sphingophospholipid and glycerophospholipids) and glycolipids with examples for each                                                                                                                                                                                                                                                                                                                 |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.BCHM.0008  | Indicate composition and functions of: Glycerophospholipids (Phosphatidylcholine, phosphatidylethanolamine, phosphatidylserine, phosphatidylinositol, cardiolipin), Sphingophospholipid (Sphingomyelin), Glycolipids (Cerebrosides, globosides and gangliosides)                                                                                                                                                                        |
|           |            |                                         | SOM.MK.1.BPM1.1.FTM.3.BCHM.0009  | Discuss role of lung surfactant (Dipalmitoyl phosphatidylcholine/ DPPC) in respiratory distress syndrome                                                                                                                                                                                                                                                                                                                                |
| Lecture 3 | PHYS       | Homeostasis                             | SOM.MK.II.BPM1.1.FTM.3.PHYS.0001 | Apply the concept of a homeostatic feedback loop to the regulation of key controlled variables (temperature, blood pressure, water content in extra- and intracellular compartments, pH) in the human body.                                                                                                                                                                                                                             |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0002 | List the typical values and normal ranges for plasma Na <sup>+</sup> , K <sup>+</sup> , H <sup>+</sup> (pH), HCO <sub>3</sub> <sup>-</sup> , Cl <sup>-</sup> , Ca <sup>+2</sup> , and glucose, and the typical intracellular pH and concentrations of Na <sup>+</sup> , K <sup>+</sup> , Cl <sup>-</sup> , Ca <sup>+2</sup> , and HCO <sub>3</sub> <sup>-</sup> .                                                                       |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0003 | Define the terms “steady state” and “equilibrium” and evaluate if example systems like specific ion and water distributions across a membrane are in either state.                                                                                                                                                                                                                                                                      |
| Lecture 4 | PHYS       | Tonicity and Osmolarity                 | SOM.MK.II.BPM1.1.FTM.3.PHYS.0007 | Define Fick's Law of diffusion, and explain how changes in the concentration gradient, surface area, time and distance will influence the diffusional movement of a compound.                                                                                                                                                                                                                                                           |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0020 | Contrast the following units used to describe concentration: mM, mEq/l, mg/dl, mg%.                                                                                                                                                                                                                                                                                                                                                     |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0021 | List the typical values and normal range for plasma osmolality. Differentiate between the terms osmolarity and osmolality.                                                                                                                                                                                                                                                                                                              |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0017 | Using a cell membrane as an example, define reflection coefficient and partition coefficient, and explain how the relative permeability of a cell to water and solutes will generate an osmotic pressure. Contrast the osmotic pressure generated across a cell membrane by a solution of particles that freely cross the membrane with that of a solution with the same osmolality, but particles that cannot cross the cell membrane. |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0022 | Describe the role of water channels (aquaporins) in facilitating the movement of water across biological membranes.                                                                                                                                                                                                                                                                                                                     |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0023 | Understand how regulation of the concentrations of K <sup>+</sup> , Cl <sup>-</sup> , and other Na <sup>+</sup> solutes influence cell volume.                                                                                                                                                                                                                                                                                          |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0024 | Predict the movement of water across a semi-permeable membrane depending on solute concentrations in both compartments.                                                                                                                                                                                                                                                                                                                 |
| Lecture 5 | PHYS       | Resting Membrane Potential              | SOM.MK.II.BPM1.1.FTM.3.PHYS.0025 | Predict the difference in free energy of a solute and solvent between two compartments, including chemical and electrical forces. Evaluate if the compartments are in a steady state and/or equilibrium.                                                                                                                                                                                                                                |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0026 | Based on the principle of ionic attraction, explain how a potential difference across a membrane will influence the distribution of a cation and an anion.                                                                                                                                                                                                                                                                              |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0027 | Write the Nernst equation and indicate how this equation accounts for both the chemical and electrical driving forces that act on an ion. Explain the effects of altering the intracellular or extracellular Na <sup>+</sup> , K <sup>+</sup> , Cl <sup>-</sup> , or Ca <sup>2+</sup> concentration on the equilibrium potential for that ion.                                                                                          |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0028 | List the concentrations and equilibrium potentials of Na, K, Cl, Ca in a typical non-excitabile cell. Based on the Nernst equilibrium potential, predict the direction of transmembrane flux for an ion when the membrane potential a) is at its equilibrium potential, b) is higher than the equilibrium potential, or c) is less than the equilibrium potential.                                                                      |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0029 | Be able to calculate the equilibrium potential for an ion using the Nernst equation. Contrast the difference in the Nernst potential for K <sup>+</sup> caused by a 5 mEq/l increase in extracellular K <sup>+</sup> with the change in the Nernst potential for Na <sup>+</sup> caused by a 5 mEq/l increase in extracellular Na <sup>+</sup> .                                                                                        |

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|-----------|------------|-----------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0030 | Using the Goldman-Hodgkin-Katz equation, explain how the relative permeabilities of the 4 key ion species (Na <sup>+</sup> , K <sup>+</sup> , Ca <sup>2+</sup> , Cl <sup>-</sup> ) create the resting membrane potential. Given an increase or decrease in the permeability for each of the ions, predict how the membrane potential would change. (Not required to calculate on exams) |
| DLA 4     | PHYS       | Epithelial Function                     | SOM.MK.II.BPM1.1.FTM.3.PHYS.0014 | Explain the functional significance of polarized distribution of ion channels and transporter proteins to the apical or basolateral cell membrane.                                                                                                                                                                                                                                      |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0018 | Draw an epithelium, labeling the tight junctions, the apical membrane, and the basolateral membrane.                                                                                                                                                                                                                                                                                    |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0349 | Trace the movement of a solute that travels across an epithelium by a transcellular pathway and a solute that travels via a paracellular pathway.                                                                                                                                                                                                                                       |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0019 | Explain the role of the tight junctions in delineating the apical and basolateral membranes.                                                                                                                                                                                                                                                                                            |
| Lecture 6 | PHYS       | Action Potentials                       | SOM.MK.II.BPM1.1.FTM.3.PHYS.0031 | Define the following properties of ion channels: gating, activation, and inactivation.                                                                                                                                                                                                                                                                                                  |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0032 | Contrast the gating of ion selective channels by extracellular ligands, intracellular ligands, stretch, and voltage.                                                                                                                                                                                                                                                                    |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0033 | Know the properties of voltage-gated Na <sup>+</sup> , K <sup>+</sup> , and Ca <sup>2+</sup> channels, and understand that voltage influences their gating, activation, and inactivation.                                                                                                                                                                                               |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0034 | Understand how the activity of voltage-gated Na <sup>+</sup> , K <sup>+</sup> , and Ca <sup>2+</sup> channels generates an action potential and the roles of those channels in each phase (depolarization, overshoot, repolarization, hyperpolarization) of the action potential.                                                                                                       |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0035 | Describe ionic basis of an action potential.                                                                                                                                                                                                                                                                                                                                            |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0036 | Draw a typical action potential and explain why afterhyperpolarization coincides with the greatest fractional K conductance but not the greatest absolute K conductance.                                                                                                                                                                                                                |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0037 | Using the Nernst equation and knowledge of sodium channel voltage dependent gating mechanisms (activation and inactivation gates), contrast the mechanisms leading to weakness and paralysis due to hypo- and hyper-kalemia.                                                                                                                                                            |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0038 | Draw a graph that shows how the voltage sensitivity of the voltage gated sodium channel changes with extracellular calcium concentration.                                                                                                                                                                                                                                               |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.3.PHYS.0039 | Contrast the role that chronic depolarization (e.g. with hyperkalemia) plays in the activation of the K and Na conductances versus rapid depolarization (graded potential towards threshold).                                                                                                                                                                                           |
| DLA 5     | ANAT       | Early Embryology                        | SOM.MK.II.BPM1.1.FTM.1.ANAT.1001 | Briefly describe the process from fertilization to the formation of the blastocyst, paying attention to the location and timeframe of each event                                                                                                                                                                                                                                        |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.1.ANAT.1002 | Describe the location and process of implantation of the blastocyst                                                                                                                                                                                                                                                                                                                     |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.1.ANAT.1003 | List the derivatives of the germ layers.                                                                                                                                                                                                                                                                                                                                                |
|           |            |                                         | SOM.MK.II.BPM1.1.MSK.1.ANAT.1022 | Describe the differentiation of the trophoblast, formation of lacunar networks and the establishment of a primordial uteroplacental circulation.                                                                                                                                                                                                                                        |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.1.ANAT.1023 | Describe the sequence of events that leads to development of the amniotic cavity, bilaminar embryo, primitive umbilical vesicle (yolk sac), extraembryonic mesoderm and its clinical significance.                                                                                                                                                                                      |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.1.ANAT.1024 | Describe the development of the notochord and its role in induction of the neural plate.                                                                                                                                                                                                                                                                                                |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.1.ANAT.1025 | Describe the fate of the intraembryonic mesoderm, including the development of the somites and intraembryonic coelom.                                                                                                                                                                                                                                                                   |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.1.ANAT.1026 | Describe the folding of the embryo, paying particular attention to the sequence and timing at which folding occurs.                                                                                                                                                                                                                                                                     |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.1.ANAT.1027 | Describe the embryology and clinical presentation of sacrococcygeal teratoma                                                                                                                                                                                                                                                                                                            |
|           |            |                                         | SOM.MK.II.BPM1.1.FTM.1.ANAT.1028 | Describe the process of gastrulation and its significance in embryonic development.                                                                                                                                                                                                                                                                                                     |
| DLA 6     | HCB        | Histology of Basic Tissues - Epithelium | SOM.MK.I.BPM1.1.FTM.3.HCB.0767   | Define epithelium.                                                                                                                                                                                                                                                                                                                                                                      |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0768   | Describe the structural and functional characteristics that distinguish epithelial tissues from the three other basic tissues.                                                                                                                                                                                                                                                          |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0769   | Recap the localization of apical specializations (cilia, flagella, primary cilia, microvilli, stereocilia)                                                                                                                                                                                                                                                                              |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0770   | Recap the description of the terminal web.                                                                                                                                                                                                                                                                                                                                              |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0771   | State the function of the glycocalyx.                                                                                                                                                                                                                                                                                                                                                   |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0772   | Describe the structure of the glycocalyx.                                                                                                                                                                                                                                                                                                                                               |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0773   | Describe the classification of epithelial tissue.                                                                                                                                                                                                                                                                                                                                       |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0774   | Differentiate between the different types of epithelia based on their structure, function and localization.                                                                                                                                                                                                                                                                             |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0775   | Discuss the basic concepts of immotile cilia syndromes in Kartagener's syndrome.                                                                                                                                                                                                                                                                                                        |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0776   | Explain the role of cell adhesion molecules (CAM) in cell adhesion                                                                                                                                                                                                                                                                                                                      |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0777   | List the families of CAM in cell-to-cell adhesion (cadherins, selectins, immunoglobulin super family, integrins)                                                                                                                                                                                                                                                                        |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0778   | List the families of CAM in cell-to-ECM adhesion (integrins)                                                                                                                                                                                                                                                                                                                            |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0779   | State the function and localization of zonula occludens.                                                                                                                                                                                                                                                                                                                                |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0780   | Describe the histological structure of zonula occludens.                                                                                                                                                                                                                                                                                                                                |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0781   | State the function and localization of zonula adherens.                                                                                                                                                                                                                                                                                                                                 |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0782   | Describe the histological structure of zonula adherens.                                                                                                                                                                                                                                                                                                                                 |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0783   | State the function and localization of gap junctions.                                                                                                                                                                                                                                                                                                                                   |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0784   | Describe the histological structure of gap junctions.                                                                                                                                                                                                                                                                                                                                   |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0785   | State the function and localization of desmosomes.                                                                                                                                                                                                                                                                                                                                      |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0786   | Describe the histological structure of desmosomes.                                                                                                                                                                                                                                                                                                                                      |

| Activity  | Discipline | Title                                   | OBJECTIVE CODE                  | Lecture Objective                                                                                                                                                            |
|-----------|------------|-----------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lecture 7 | HCB        | Histology of Epithelium & Glands        | SOM.MK.I.BPM1.1.FTM.3.HCB.0787  | Describe the terminal bar.                                                                                                                                                   |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0788  | State the function and localization of hemidesmosomes.                                                                                                                       |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0789  | Describe the histological structure of hemidesmosomes.                                                                                                                       |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0790  | State the function and localization of focal adhesions.                                                                                                                      |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0791  | Describe the histological structure of focal adhesions.                                                                                                                      |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0792  | Give function(s) of lateral digitations.                                                                                                                                     |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0793  | Describe the structure of lateral digitations.                                                                                                                               |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0794  | Identify lateral digitations on micrographs.                                                                                                                                 |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0795  | Give function(s) of basal infoldings.                                                                                                                                        |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0796  | Describe the structure of basal infoldings.                                                                                                                                  |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0797  | Identify basal infoldings on micrographs.                                                                                                                                    |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0798  | Give the function of the basal lamina.                                                                                                                                       |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0799  | Describe the microscopic structure of the basal lamina.                                                                                                                      |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0800  | Identify the basal lamina on micrographs.                                                                                                                                    |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0801  | Describe the microscopic structure of the basement membrane.                                                                                                                 |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0802  | Compare and contrast basement membrane vs. basal lamina vs external lamina.                                                                                                  |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0803  | Compare and contrast exocrine vs. endocrine glands.                                                                                                                          |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0804  | Distinguish between unicellular and multicellular exocrine glands.                                                                                                           |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0805  | Describe the three modes of secretion in multicellular exocrine glands.                                                                                                      |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0806  | Describe the classification of multicellular exocrine glands.                                                                                                                |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0807  | Describe the microscopic appearance of serous glands.                                                                                                                        |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0808  | Describe the microscopic appearance of mucous glands.                                                                                                                        |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0809  | Describe the microscopic appearance of mixed glands.                                                                                                                         |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0810  | Describe the histology of a compound salivary gland.                                                                                                                         |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0811  | Identify parotid, submandibular and sublingual salivary glands on photomicrographs.                                                                                          |
| DLA 7     | BCHM       | Structure and Function of Nucleic Acids | SOM.MK.I.BPM1.1.FTM.3.BCHM.0020 | Distinguish between purines, pyrimidines; ribonucleosides, deoxyribonucleosides; ribonucleotides and deoxyribonucleotides.                                                   |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0021 | Describe the phosphodiester bond                                                                                                                                             |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0022 | Describe the three important features of the DNA double helix (complementary, antiparallel, and forces stabilizing the helix).                                               |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0023 | Discuss the physicochemical properties of DNA                                                                                                                                |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0024 | Describe the basic structures and roles of the major classes of RNA (mRNA, tRNA, rRNA, snRNA, miRNA)                                                                         |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0025 | Predict the sequence of the complementary strand if the sequence of one of the DNA strands is provided.                                                                      |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0026 | Analyze the nucleotide composition of double stranded DNA using the Chargaff's rule                                                                                          |
| DLA 8     | BCHM       | DNA Packaging                           | SOM.MK.I.BPM1.1.FTM.3.BCHM.0028 | Differentiate between a supercoiled and a relaxed DNA molecule                                                                                                               |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0029 | State the reasons why supercoiling of DNA molecules is important.                                                                                                            |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0030 | Compare and contrast DNA packaging in prokaryotes and eukaryotes.                                                                                                            |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0031 | Describe the structure of chromatin and histone proteins.                                                                                                                    |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0032 | Explain how dsDNA and histones associate to form the nucleosome.                                                                                                             |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0033 | Describe the different structural conformations of DNA packaging in eukaryotes (10 nm Fiber , 30 nm fiber and metaphase chromosomes).                                        |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0034 | Describe how histone modification can influence transcriptional activity                                                                                                     |
| DLA 9     | BCHM       | DNA Replication and Telomerase          | SOM.MK.I.BPM1.1.FTM.3.BCHM.0048 | Describe the role of telomeres and telomerase in eukaryotic DNA replication                                                                                                  |
| Lecture 8 | BCHM       | DNA Replication                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0035 | Describe semi-conservative replication of DNA                                                                                                                                |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0036 | Compare and contrast the similarities and differences between DNA Pol I and DNA Pol III of E. coli                                                                           |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0037 | Describe how DNA is synthesized from its 5' to 3' end                                                                                                                        |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0038 | Explain the significance of the origin of replication during the process of DNA synthesis.                                                                                   |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0039 | Explain the problems associated with unwinding of the DNA double helix                                                                                                       |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0040 | Describe the role of helicase enzymes in unwinding DNA prior to its replication                                                                                              |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0041 | Describe the role of single-strand binding protein & topoisomerases in DNA replication                                                                                       |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0042 | Outline the order of events from RNA priming to completed DNA during replication in E.coli.                                                                                  |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0043 | Distinguish between synthesis of leading and lagging strands of DNA. Identify the significance of DNA ligase in lagging strand synthesis                                     |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0044 | Describe the significance of the various proteins and enzymes required for replication and predict the effects of mutations of these proteins and enzymes on DNA replication |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0045 | Evaluate the proofreading activity of DNA polymerase                                                                                                                         |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0046 | Differentiate between DNA replication in eukaryotes and prokaryotes                                                                                                          |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0047 | Outline the mechanisms of action of drugs that interfere with DNA replication                                                                                                |
| DLA 10    | BCHM       | RNA Polymerase and Transcription        | SOM.MK.I.BPM1.1.FTM.3.BCHM.0049 | Describe the structural features of a gene and their functions                                                                                                               |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0050 | Describe the types and functions of the different cellular RNAs (mRNA, tRNA, rRNA, snRNA, miRNA)                                                                             |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0051 | Describe the major steps of RNA synthesis in prokaryotes and eukaryotes                                                                                                      |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0052 | Explain the function and structure of DNA sequences that are involved in the initiation (promoters) and termination of transcription                                         |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0053 | Predict the sequence of the transcribed RNA if the template DNA strand or the coding strand of the RNA is provided                                                           |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0054 | List the RNA products that are transcribed by the three different types of RNA polymerases in eukaryotes                                                                     |
|           |            |                                         | SOM.MK.I.BPM1.1.FTM.3.BCHM.0057 | Explain the mechanism of action of rifampin and alpha-amanitin as inhibitors of transcription.                                                                               |



| Activity   | Discipline | Title                                           | OBJECTIVE CODE                    | Lecture Objective                                                                                                                                                                                                                                                                                                                        |
|------------|------------|-------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0058   | Describe the effect of the antibiotic drugs that interfere with DNA gyrase such as (quinolones) ciprofloxacin                                                                                                                                                                                                                            |
| Lecture 9  | BCHM       | Post-transcriptional Modifications              | SOM.MK.I.BPM1.1.FTM.3.BCHM.0059   | Differentiate between the structures of prokaryotic and eukaryotic mRNAs                                                                                                                                                                                                                                                                 |
|            |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0060   | Describe the post-transcriptional modifications of eukaryote mRNA and the significance of these modifications (5' capping, polyadenylation, and splicing)                                                                                                                                                                                |
|            |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0061   | Predict effects of splice site mutations on mature mRNA and ultimately the translated protein                                                                                                                                                                                                                                            |
|            |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0062   | Describe how RNA editing affects our understanding of transmission of genetic information from the genome to protein                                                                                                                                                                                                                     |
|            |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0063   | Describe human disease conditions that may be associated with mRNA modification and errors in this process                                                                                                                                                                                                                               |
| Lecture 10 | GNET       | Mitosis, Meiosis and Non-disjunction            | SOM.MK.III.BPM1.1.FTM.3.GNET.0001 | Recognize differences in rates of cell division in different tissues                                                                                                                                                                                                                                                                     |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0002 | Analyze stages of the cell cycle - (G <sub>1</sub> , G <sub>0</sub> , S, G <sub>2</sub> , M)                                                                                                                                                                                                                                             |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0003 | Indicate the role of p53 in the cell cycle: Role in genome surveillance and its role in initiation of apoptosis                                                                                                                                                                                                                          |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0004 | Describe the chromosomal structure changes in different phases of the cell cycle                                                                                                                                                                                                                                                         |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0005 | Explain the significance of mitosis, summarize the events, and analyze structures of mitosis including Prophase: Events in the nucleus, formation of spindle in cytoplasm, Metaphase: Association of chromosomes with spindle fibres, Anaphase: Separation of sister chromatids, Telophase: Cytokinesis, re-formation of nuclei          |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0006 | Recognize the significance of meiosis and summarize its main achievements                                                                                                                                                                                                                                                                |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0007 | Describe the sequence of events that comprise meiosis.                                                                                                                                                                                                                                                                                   |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0008 | Analyze the effect of a nondisjunction of the sex chromosomes in meiosis 1 or 2 during gametogenesis which lead to: Turner, Klinefelter, XYY, or XXX syndromes.                                                                                                                                                                          |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0009 | Interpret the significance of recombination in prophase I of meiosis. Identify the role of meiosis in gametogenesis. Analyze events and structures that occur in meiosis                                                                                                                                                                 |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0010 | Compare and contrast the overall features of mitosis with meiosis and explain how meiosis introduces diversity into the genotype of the offspring after sexual reproduction                                                                                                                                                              |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0011 | Describe the distinguishing features and the morphological classification of chromosomes including metacentric, submetacentric, acrocentric, and telocentric                                                                                                                                                                             |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0012 | Outline the differences between meiosis in males and females                                                                                                                                                                                                                                                                             |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0013 | Compare and contrast the outcome of nondisjunction that might occur during meiosis or in mitosis                                                                                                                                                                                                                                         |
| Lecture 11 | GNET       | Genomics and Human Variation                    | SOM.MK.III.BPM1.1.FTM.3.GNET.0016 | Compare and contrast different types of repetitive DNA including low copy repeats, SSR, VNTR, LINE, SINE, pseudogenes, and gene families                                                                                                                                                                                                 |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0017 | Compare and contrast single nucleotide polymorphisms to rare variants                                                                                                                                                                                                                                                                    |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0018 | Discuss the structure of chromosomes in the human (male and female), and general terminology used to describe karyotypes; Compare nuclear DNA to organellar DNA (such as the mitochondrial                                                                                                                                               |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0019 | Analyze dosage compensation by X-chromosome inactivation                                                                                                                                                                                                                                                                                 |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0105 | Explain the normal function of epigenetic gene silencing in development; including the role of X-inactivation as discussed Genomics and Human Variation                                                                                                                                                                                  |
| DLA 11     | GNET       | Inheritance: AD and AR and Pseudo Dominance     | SOM.MK.III.BPM1.1.FTM.3.GNET.0020 | Describe basic definitions of human genetics including: locus, gene, allele, genotype, and zygosity                                                                                                                                                                                                                                      |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0021 | From a description of a pedigree, identify autosomal dominant (AD) and autosomal recessive (AR) inheritance; Compare and contrast Mendelian and non-Mendelian genetics                                                                                                                                                                   |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0022 | Calculate recurrence risk of autosomal dominant and autosomal recessive disorders within families                                                                                                                                                                                                                                        |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0023 | Analyze family pedigree to determine the risk that immediate family members might be carriers of disease alleles (AR or X-linked) and when to use the 2/3 rule                                                                                                                                                                           |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0024 | Describe and differentiate between pseudoautosomal dominance, co-dominance, and incomplete dominance; give examples                                                                                                                                                                                                                      |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0025 | Describe co-dominance, giving examples                                                                                                                                                                                                                                                                                                   |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0026 | Describe incomplete dominance, giving examples                                                                                                                                                                                                                                                                                           |
| Lecture 12 | GNET       | Flipped Class: Mendelian Inheritance: AD and AR | SOM.MK.III.BPM1.1.FTM.3.GNET.0027 | Interpret a family pedigree incorporating standard genetic symbols. Interpret a pedigree and identify features, disorders, genotype, carrier status, recurrence risk, and genetic principles                                                                                                                                             |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0028 | Demonstrate how disease liability is transmitted for autosomal dominant and autosomal recessive disorders, beginning from the first "de novo" occurrence                                                                                                                                                                                 |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0029 | Propose how the basic risk calculations are performed for the different categories of genetic transmission: AR and AD                                                                                                                                                                                                                    |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0030 | Identify obligate carriers in a pedigree: AR, or AD with reduced penetrance                                                                                                                                                                                                                                                              |
|            |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0031 | Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following autosomal dominant disorders: Huntington disease, myotonic dystrophy, familial hypercholesterolemia, osteogenesis imperfecta, Marfan syndrome, Achondroplasia, NF, acute intermittent porphyria) |

| Activity   | Discipline | Title                               | OBJECTIVE CODE                   | Lecture Objective                                                                                                                                                                                                                                                                                                                  |
|------------|------------|-------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |            | Inheritance: AD and AR              | SOM.MKIII.BPM1.1.FTM.3.GNET.0032 | Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following following autosomal recessive disorders: CF, PKU, Tay-Sachs disease, Sickle cell disease, thalassemia, hemochromatosis, homocystineuria, SCID due to ADA deficiency, alkaptonuria          |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0033 | Identify pseudodominant inheritance and indicate factors that result in this inheritance pattern. Be able to identify this concept from clinical scenarios and pedigrees                                                                                                                                                           |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0034 | Compare and contrast locus heterogeneity and allelic heterogeneity; give examples                                                                                                                                                                                                                                                  |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0035 | Explain how allelic heterogeneity allows the formation of a compound heterozygote in autosomal recessive disorders                                                                                                                                                                                                                 |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0036 | Define the terms and give examples for: haploinsufficiency, dominant negative effect, gain of function mutations, loss of function mutations                                                                                                                                                                                       |
| DLA 12     | GNET       | Inheritance: X-linkage              | SOM.MKIII.BPM1.1.FTM.3.GNET.0037 | Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following X-linked disorders: Hemophilia A and B, G6PD deficiency, Duchenne and Becker muscular dystrophy, red/green color blindness. Lesch Nyhan syndrome, X-SCID as examples of X-linked disorders |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0038 | Summarize and distinguish the characteristics of the following patterns of inheritance using pedigrees: autosomal dominant, autosomal recessive, X-linked dominant and X-linked recessive                                                                                                                                          |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0039 | Demonstrate how disease liability is transmitted for X-linked diseases                                                                                                                                                                                                                                                             |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0040 | Propose how the basic risk calculations are performed for X-linked disorders                                                                                                                                                                                                                                                       |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0041 | Explain how a woman who is a carrier of an X-linked disorder might manifest symptoms of the condition (manifesting heterozygote) Skewed X-inactivation, non-random X-inactivation                                                                                                                                                  |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0042 | Discuss X-linked dominant disorders and traits                                                                                                                                                                                                                                                                                     |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0043 | Compare and contrast variable expressivity and incomplete penetrance; Give examples of each                                                                                                                                                                                                                                        |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0044 | Describe pleiotropy                                                                                                                                                                                                                                                                                                                |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0045 | Compare and contrast locus heterogeneity and allelic heterogeneity; gve examples                                                                                                                                                                                                                                                   |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0046 | Describe locus heterogeneity in terms of complementation groups                                                                                                                                                                                                                                                                    |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0047 | Explain how allelic heterogeneity allows the formation of a compound heterozygote in autosomal recessive disorders                                                                                                                                                                                                                 |
| Lecture 13 | GNET       | Flipped Class: X-Linked Inheritance | SOM.MKIII.BPM1.1.FTM.3.GNET.0048 | Demonstrate how disease liability is transmitted for autosomal dominant diseases, autosomal recessive diseases and X-linked diseases                                                                                                                                                                                               |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0049 | Distinguish the different characteristics of recessive and dominant X-linked traits using pedigrees                                                                                                                                                                                                                                |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0050 | Identify obligate carriers in a pedigree depicting an X-linked trait                                                                                                                                                                                                                                                               |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0051 | Compare and contrast between X-linked, Y-linked, AR, AD, and mitochondrial traits.                                                                                                                                                                                                                                                 |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0052 | Discuss the concept of penetrance and variable expression and Differentiate between locus and allelic heterogeneity                                                                                                                                                                                                                |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0053 | Calculate the recurrence risk based on penetrance for autosomal dominant and autosomal recessive disorders                                                                                                                                                                                                                         |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0054 | Define pleiotropy with examples. Be able to identify concepts if given a clinical scenario                                                                                                                                                                                                                                         |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0055 | Recognize occurrence of new mutations as a cause of new genetic diseases in families. Be able to identify this concept from clinical scenarios and pedigrees                                                                                                                                                                       |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0056 | Explain the phenomenon of germinal (germ-line) mosaicism                                                                                                                                                                                                                                                                           |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0057 | Identify genetic disorders that have a delayed age of onset, and be able to identify this concept from clinical scenarios and pedigrees                                                                                                                                                                                            |
|            |            |                                     | SOM.MKIII.BPM1.1.FTM.3.GNET.0058 | Explain the phenomenon of a manifesting heterozygote in X-linked recessive disorders, and be able to identify this concept from clinical scenarios and pedigrees                                                                                                                                                                   |
| DLA 13     | BCHM       | Genetic code and Translation        | SOM.MK.I.BPM1.1.FTM.3.BCHM.0064  | Explain why the genetic code is collinear with DNA, triplet in nature, redundant (degenerate), and non-overlapping.                                                                                                                                                                                                                |
|            |            |                                     | SOM.MK.I.BPM1.1.FTM.3.BCHM.0065  | Define the terms: reading frame and frame-shift mutation.                                                                                                                                                                                                                                                                          |
|            |            |                                     | SOM.MK.I.BPM1.1.FTM.3.BCHM.0066  | Interpret the standard genetic code table. Identify the initiation codon and the three termination codons.                                                                                                                                                                                                                         |
|            |            |                                     | SOM.MK.I.BPM1.1.FTM.3.BCHM.0067  | Explain the universal nature of the genetic code (with minor differences in mitochondria) and predict the sequence of a peptide if the coding strand of the DNA is provided.                                                                                                                                                       |
|            |            |                                     | SOM.MK.I.BPM1.1.FTM.3.BCHM.0068  | Discuss how protein synthesis proceeds from the N-terminal to their C-terminal ends and explain how this creates a naming convention for peptides.                                                                                                                                                                                 |
|            |            |                                     | SOM.MK.I.BPM1.1.FTM.3.BCHM.0050  | List and describe the major function of the following types of RNA in cells: mRNA, tRNA, rRNA, snRNA, miRNA.                                                                                                                                                                                                                       |
|            |            |                                     | SOM.MK.I.BPM1.1.FTM.3.BCHM.0070  | Describe the subunit composition prokaryotic (70S) and eukaryotic (80S) ribosomes. Recognize the roles of A, P, and E sites on the ribosome.                                                                                                                                                                                       |
|            |            |                                     | SOM.MK.I.BPM1.1.FTM.3.BCHM.0071  | Describe the structure, function, and charging of tRNAs.                                                                                                                                                                                                                                                                           |
|            |            |                                     | SOM.MK.I.BPM1.1.FTM.3.BCHM.0072  | Describe the codon/anticodon interaction and discuss the wobble hypothesis.                                                                                                                                                                                                                                                        |
|            |            |                                     | SOM.MK.I.BPM1.1.FTM.3.BCHM.0073  | Describe the sequence of events that occurs during translation in prokaryotes (initiation, elongation and termination) and list the major differences between prokaryotic and eukaryotic translation.                                                                                                                              |
|            |            |                                     | SOM.MK.I.BPM1.1.FTM.3.BCHM.0074  | Explain how miRNA and diphtheria toxin interfere with eukaryotic translation.                                                                                                                                                                                                                                                      |

| Activity           | Discipline | Title                                           | OBJECTIVE CODE                    | Lecture Objective                                                                                                                                                                                                                                                                                                   |
|--------------------|------------|-------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lecture 14         | BCHM       | Translation and Post-translational Modification | SOM.MK.I.BPM1.1.FTM.3.BCHM.0069   | Explain the mode of action of common antibiotics that interfere with translation: Initiation inhibitors (streptomycin), Elongation inhibitors (Tetracycline, chloramphenicol, erythromycin, puromycin, cycloheximide).                                                                                              |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0075   | Review examples of post-translational modifications: Zymogen activation (trypsinogen to trypsin), Serine/threonine phosphorylation (regulation of enzymes in metabolism), Tyrosine phosphorylation (Insulin receptor), O-linked glycosylation, N-linked glycosylation and Lipid anchoring (farnesyl groups to Ras). |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0076   | Describe proteolytic processing and post-translational modifications using insulin as an example.                                                                                                                                                                                                                   |
| DLA 14             | BCHM       | Amino Acids: Structure & Function               | SOM.MK.I.BPM1.1.FTM.3.BCHM.0078   | Classify amino acids based on their R groups (with examples for each group).                                                                                                                                                                                                                                        |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0079   | Name the amino acids with a charged side chain, the amino acids containing a hydroxyl group and the amino acids containing sulphur.                                                                                                                                                                                 |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0081   | Discuss the role of histidine residues in buffering in hemoglobin.                                                                                                                                                                                                                                                  |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0083   | Discuss the importance of derivatives of amino acids (GABA, histamine, serotonin, nitric oxide, and catecholamines).                                                                                                                                                                                                |
| Lecture 15         | BCHM       | Structure and Function of Proteins              | SOM.MK.I.BPM1.1.FTM.3.BCHM.0087   | Differentiate between the hierarchical levels of protein structure using the example of myoglobin and hemoglobin.                                                                                                                                                                                                   |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0088   | Discuss the significance of bond forces involved in maintenance of protein structure (peptide bonds, disulfide bonds, hydrogen bonds, ionic bonds, hydrophobic forces and Van der Waals forces) using hemoglobin as an example.                                                                                     |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0090   | Relate that missense mutations can alter the solubility, function, charge and electrophoretic properties of proteins.                                                                                                                                                                                               |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0091   | Discuss neurochemical abnormalities in Prion disease and Alzheimer disease.                                                                                                                                                                                                                                         |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0092   | Describe the role of chaperones and ubiquitin.                                                                                                                                                                                                                                                                      |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0093   | Discuss protein denaturation in vivo and in the laboratory.                                                                                                                                                                                                                                                         |
| Lecture 16         | GNET       | Unexpected Events in Inheritance                | SOM.MK.III.BPM1.1.FTM.3.GNET.0059 | Explain how the classic pattern of inheritance may be disrupted by incomplete penetrance, variable expressivity and variable age-of-onset.                                                                                                                                                                          |
|                    |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0060 | Describe examples of genetic abnormalities that may result in a non-Mendelian pattern of inheritance                                                                                                                                                                                                                |
|                    |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0061 | Discuss disorders due to mutations in mitochondrial genes: Leber hereditary optic neuropathy (LHON), MELAS, MERRF                                                                                                                                                                                                   |
|                    |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0062 | Identify the phenomenon of heteroplasmy in mitochondrial disorders and indicate its genetic basis                                                                                                                                                                                                                   |
|                    |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0063 | Predict the effect of imprinting in the transmission of genetic disease                                                                                                                                                                                                                                             |
|                    |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0064 | Identify imprinting defects (Prader Willi syndrome, Angelman syndrome)                                                                                                                                                                                                                                              |
|                    |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0065 | Explain the phenomenon of uniparental disomy and microdeletion in Prader-Willi and Angelman syndrome                                                                                                                                                                                                                |
|                    |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0066 | Analyze how triplet repeat expansion mutations result in anticipation (non-Mendelian inheritance) for Huntington disease, myotonic dystrophy, Fragile X syndrome.                                                                                                                                                   |
|                    |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0067 | Explain digenic inheritance using retinitis pigmentosa as an example                                                                                                                                                                                                                                                |
|                    |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0068 | Determine a strategy for identifying Non-Mendelian disorders, investigating them and providing appropriate genetic counselling for families                                                                                                                                                                         |
|                    |            |                                                 | SOM.MK.III.BPM1.1.FTM.3.GNET.0110 | Demonstrate how genomic imprinting would appear in a human pedigree when either a male or female passes a mutation to offspring                                                                                                                                                                                     |
| Reading Assignment | GNET       | Molecular Biology Techniques                    | SOM.MK.I.BPM1.1.FTM.3.GNET.0200   | Explain the general principles and utilization of various molecular biology tools and techniques, including Restriction endonucleases, DNA cloning, Probes, Southern Blotting, Restriction Fragment Length Polymorphisms, Polymerase chain reaction and Gene Expression analysis                                    |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.GNET.0201   | Summarize how complementary DNA (cDNA) is generated from messenger RNA (mRNA) and how this technique is used to express human proteins in a bacteria to generate therapeutic proteins such as insulin                                                                                                               |
| Lecture 17         | GNET       | Molecular Biology Techniques                    | SOM.MK.I.BPM1.1.FTM.3.GNET.0202   | Identify the differences between mutations which affect the structures of a gene (or gene region), the structure of the transcribed mRNA product and the structure of the translated protein                                                                                                                        |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.GNET.0203   | Explain the concept of hybridization: between a complementary single stranded DNA probe and its target (DNA or RNA); between an antibody and its protein target                                                                                                                                                     |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.GNET.0204   | Compare data from RFLP, ASO, DNA sequencing: from normal homozygotes, carriers (heterozygotes), and affected homozygotes                                                                                                                                                                                            |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.GNET.0205   | Predict the application of molecular biology techniques for detection of a single base mutation versus small insertion or deletion of DNA sequence, versus large insertion or deletion of DNA sequence.                                                                                                             |
| Reading Assignment | BCHM       | Properties of Enzymes                           | SOM.MK.I.BPM1.1.FTM.3.BCHM.0103   | Describe the factors that influence enzyme reaction rates (substrate, pH and temperature).                                                                                                                                                                                                                          |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0104   | Discuss enzyme regulation via the concentrations of substrates or products.                                                                                                                                                                                                                                         |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0105   | Describe enzyme regulation via the modulation of enzyme concentrations.                                                                                                                                                                                                                                             |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0106   | Discuss conformation of enzymes and changes of conformation in relation to enzyme regulation. (reversible covalent modification, proteolytic cleavage and allosteric regulation).                                                                                                                                   |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0107   | Describe the concept of proteolytic activation of digestive enzymes (eg: zymogens such as trypsinogen).                                                                                                                                                                                                             |
|                    |            |                                                 | SOM.MK.I.BPM1.1.FTM.3.BCHM.0108   | Discuss the kinetics of allosteric enzymes and explain the sigmoidal graph.                                                                                                                                                                                                                                         |

| Activity   | Discipline | Title                                      | OBJECTIVE CODE                    | Lecture Objective                                                                                                                                                                                                           |
|------------|------------|--------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lecture 18 | BCHM       | Enzymes                                    | SOM.MK.I.BPM1.1.FTM.3.BCHM.0109   | Describe allosteric enzymes and their regulation related to changes of $K_{0.5}$ or $V_{max}$ . Discuss in general the regulation of metabolic pathways related to feedback inhibition and feed-forward activation.         |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0110   | Describe Michaelis-Menten kinetics                                                                                                                                                                                          |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0111   | Define $K_m$ and $V_{max}$ and discuss their significance with the help of a graph.                                                                                                                                         |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0112   | Discuss the inhibition caused by statin drugs (enzyme inhibited, effect on kinetics).                                                                                                                                       |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0113   | Compare and contrast the mode of action and kinetics of reversible competitive and noncompetitive inhibitors.                                                                                                               |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0114   | Compare and contrast the Michaelis-Menten and Lineweaver-Burk plots.                                                                                                                                                        |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0115   | Indicate X- and Y- intercepts in the Lineweaver-Burk plot                                                                                                                                                                   |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0116   | Compare and contrast competitive and noncompetitive inhibition using Michaelis-Menten graph and Lineweaver-Burk plot.                                                                                                       |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0117   | Distinguish between reversible and irreversible inhibitors.                                                                                                                                                                 |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0118   | Discuss inhibition caused by DFP and aspirin (enzymes inhibited, effects of inhibition).                                                                                                                                    |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0119   | Discuss suicide inhibition of enzymes using allopurinol as example.                                                                                                                                                         |
| DLA 15     | GNET       | Genomics in Diagnosis                      | SOM.MK.III.BPM1.1.FTM.3.GNET.0069 | Explain how triplet repeat expansion disorders are diagnosed                                                                                                                                                                |
|            |            |                                            | SOM.MK.III.BPM1.1.FTM.3.GNET.0070 | Discuss why each triplet repeat expansion disorder needs its own unique test for diagnosis                                                                                                                                  |
|            |            |                                            | SOM.MK.III.BPM1.1.FTM.3.GNET.0071 | Explain how a G-band karyotype is prepared                                                                                                                                                                                  |
|            |            |                                            | SOM.MK.III.BPM1.1.FTM.3.GNET.0072 | Describe FISH and how it can be used in diagnostic applications                                                                                                                                                             |
|            |            |                                            | SOM.MK.III.BPM1.1.FTM.3.GNET.0073 | Compare and contrast : Metaphase FISH, Interphase FISH, and Chromosome painting (SKY FISH, or spectral karyotype)                                                                                                           |
|            |            |                                            | SOM.MK.III.BPM1.1.FTM.3.GNET.0074 | Describe the strategy behind the microarray: CGH Microarray, SNP Microarray, cDNA Microarray                                                                                                                                |
| Lecture 19 | GNET       | Use of Genomics in Diagnosis               | SOM.MK.III.BPM1.1.FTM.3.GNET.0075 | Analyze genomic approaches that are used in the diagnosis of human disorders                                                                                                                                                |
|            |            |                                            | SOM.MK.III.BPM1.1.FTM.3.GNET.0076 | Analyze how PCR can be used to obtain a diagnosis of a genetic disorder                                                                                                                                                     |
|            |            |                                            | SOM.MK.III.BPM1.1.FTM.3.GNET.0077 | Discuss how next generation sequencing is being used in genomic applications                                                                                                                                                |
|            |            |                                            | SOM.MK.III.BPM1.1.FTM.3.GNET.0078 | Describe the difference between whole genome sequencing (WGS), whole exome sequencing (WES), gene panels, and single targeted gene sequencing                                                                               |
|            |            |                                            | SOM.MK.III.BPM1.1.FTM.3.GNET.0079 | Describe the different types of mutations that may be observed in the human genome and explain how they are categorized.                                                                                                    |
|            |            |                                            | SOM.MK.III.BPM1.1.FTM.3.GNET.0080 | Consider the relative pros and cons of the genomic and genetic techniques described in this section, and their limits of resolution                                                                                         |
| Lecture 20 | BCHM       | Signal Transduction                        | SOM.MK.I.BPM1.1.FTM.3.BCHM.0120   | Describe different types of cell signalling: endocrine, autocrine, paracrine, neuronal & contact-dependent.                                                                                                                 |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0121   | Indicate four different classes of receptors (steroid receptors, ion-channel receptor, receptor enzyme, receptors that form second messengers including G-protein-coupled receptors) with specific examples for each class. |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0122   | Discuss events involved in signal transduction                                                                                                                                                                              |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0123   | Identify the location of the receptors for steroid hormones. Describe intracellular signal transduction by steroid hormones.                                                                                                |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0124   | Describe the molecular changes when insulin binds to its receptor.                                                                                                                                                          |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0125   | Describe the different target (effector) enzymes of G proteins.                                                                                                                                                             |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0126   | Describe the two major second messenger systems (adenylate cyclase and phosphoinositide systems) associated with G-proteins and signal termination.                                                                         |
| Lecture 21 | BCHM       | Second Messenger Systems                   | SOM.MK.I.BPM1.1.FTM.3.BCHM.0127   | Explain how cholera toxin modifies $G_s$ and how pertussis toxin modifies $G_i$ . Predict the effects of these modifications in the respective cells.                                                                       |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0128   | Discuss the roles of cAMP, IP <sub>3</sub> , DAG and Calcium as mediators of signal transduction in different cell types.                                                                                                   |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0129   | Review the mechanism of action of phosphodiesterase inhibitors.                                                                                                                                                             |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0130   | Describe activation of adenylate cyclase by $G_s$ alpha and phospholipase C by $G_q$ -alpha subunits.                                                                                                                       |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0131   | Describe the formation and biological role of cGMP. Indicate significance of guanylate cyclase.                                                                                                                             |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0132   | Identify mechanism of action of nitric oxide (nitroglycerin) and its role in smooth muscle relaxation and vasodilatation.                                                                                                   |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0133   | Compare the cellular activation of protein kinases A, C and G.                                                                                                                                                              |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0134   | Distinguish signaling mechanisms associated with alpha-1, alpha-2 and beta-adrenergic receptors, glucagon receptors and insulin receptors.                                                                                  |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0135   | Predict the changes when an agonist or antagonist of a specific signaling pathway is administered.                                                                                                                          |
| Lecture 22 | GNET       | Flipped Class: Genetics and Genomics Cases | SOM.MK.III.BPM1.1.FTM.3.GNET.0081 | Analyze examples of molecular diagnosis obtained from genetic and genomic laboratory analysis                                                                                                                               |
| Lecture 23 | BCHM       | Clinical Enzymology                        | SOM.MK.I.BPM1.1.FTM.3.BCHM.0136   | Review the significance of estimation of LDH.                                                                                                                                                                               |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0137   | Differentiate isozymes of creatine kinase based on tissue location and subunit composition.                                                                                                                                 |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0138   | Discuss utility of serum injury markers following myocardial infarction: Myoglobin, CK-MB/total CK ratio, serum cardiac troponins I and T                                                                                   |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0139   | Discuss the utility of serum injury markers following MI: Time frame of serum injury markers (myoglobin, cardiac troponins, CK-MB) with the help of a graph.                                                                |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0140   | Describe the application of ALT, AST, ALP and GGT as markers of hepatocellular disease, biliary disease and alcohol liver disease.                                                                                          |
|            |            |                                            | SOM.MK.I.BPM1.1.FTM.3.BCHM.0141   | Discuss significance of serum amylase and serum lipase related to pancreatic disease.                                                                                                                                       |



| Activity   | Discipline | Title                                       | OBJECTIVE CODE                  | Lecture Objective                                                                                                                                                                                                    |
|------------|------------|---------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.BCHM.0142 | Indicate specific serum injury markers for bone disease (ALP), prostate cancer (PSA) and liver cancer (alpha-fetoprotein).                                                                                           |
| Lecture 24 | HCB        | Apoptosis and Necrosis                      | SOM.MK.I.BPM1.1.FTM.3.HCB.0819  | Define apoptosis                                                                                                                                                                                                     |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0820  | Define necrosis                                                                                                                                                                                                      |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0821  | State the importance of apoptosis                                                                                                                                                                                    |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0822  | State the importance of necrosis                                                                                                                                                                                     |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0823  | Discuss common causes of cellular injury.                                                                                                                                                                            |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0824  | Compare reversible vs. irreversible injuries.                                                                                                                                                                        |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0825  | Describe the early cellular responses to injury.                                                                                                                                                                     |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0826  | Discuss some examples of physiological processes that lead to cell death by apoptosis                                                                                                                                |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0827  | Discuss some examples of pathological processes that lead to cell death by necrosis                                                                                                                                  |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0828  | Discuss the morphological changes associated with cell death by apoptosis.                                                                                                                                           |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0829  | Discuss the morphological changes associated with cell death by necrosis.                                                                                                                                            |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0830  | Discuss the biochemical features associated with cell death by apoptosis.                                                                                                                                            |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0831  | Discuss the biochemical features associated with cell death by necrosis.                                                                                                                                             |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0832  | Describe the intrinsic apoptosis pathway.                                                                                                                                                                            |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0833  | Describe the extrinsic apoptosis pathway.                                                                                                                                                                            |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0834  | Describe the role of caspases in apoptosis.                                                                                                                                                                          |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0835  | Discuss methods to assess apoptosis.                                                                                                                                                                                 |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0836  | Discuss methods to assess necrosis.                                                                                                                                                                                  |
| Lecture 25 | HCB        | Cellular Adaptation                         | SOM.MK.I.BPM1.1.FTM.3.HCB.0837  | Define the term "hypertrophy" and list examples.                                                                                                                                                                     |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0838  | Define the term "hyperplasia" and list examples.                                                                                                                                                                     |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0839  | Define the term "atrophy" and list examples.                                                                                                                                                                         |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0840  | Define the term "involution" and list example(s).                                                                                                                                                                    |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0841  | Define the term "metaplasia" and list examples.                                                                                                                                                                      |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0842  | Define the terms "failure of differentiation" and "cellular atypia".                                                                                                                                                 |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0843  | Define the term "dysplasia" and list examples.                                                                                                                                                                       |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0844  | Define the term "neoplasia" and list examples.                                                                                                                                                                       |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0845  | Compare and contrast benign vs. malignant neoplasms.                                                                                                                                                                 |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0846  | Define the terms "carcinoma in situ" and "invasion".                                                                                                                                                                 |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0847  | Describe the classification of cancer cells including carcinoma, sarcoma and leukemia.                                                                                                                               |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0848  | Discuss the hallmarks of cancer.                                                                                                                                                                                     |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0849  | Describe tumor growth and progression.                                                                                                                                                                               |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0850  | Define the term "primary tumor".                                                                                                                                                                                     |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0851  | Define the term "metastases" (secondary tumors).                                                                                                                                                                     |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.HCB.0852  | List the main modes for spread of malignant neoplasms.                                                                                                                                                               |
| Lecture 26 | BCHM       | Molecular Mechanisms in Inherited Disorders | SOM.MK.I.BPM1.1.FTM.3.BCHM.0179 | Describe the molecular basis for clinical features, diagnostic tests and management of Cystic Fibrosis                                                                                                               |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.BCHM.0180 | Describe the molecular basis for clinical features, diagnostic tests and management of Sickle cell anemia                                                                                                            |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.BCHM.0181 | Describe the molecular basis for clinical features and diagnostic tests in Duchenne Muscular Dystrophy and Becker Muscular Dystrophy                                                                                 |
|            |            |                                             | SOM.MK.I.BPM1.1.FTM.3.BCHM.0182 | Compare the differences between Duchenne muscular dystrophy and Becker muscular dystrophy with reference to severity of the disorder, type of mutations, immunohistochemistry findings and alterations in dystrophin |